

Q 3.

(SOSO)

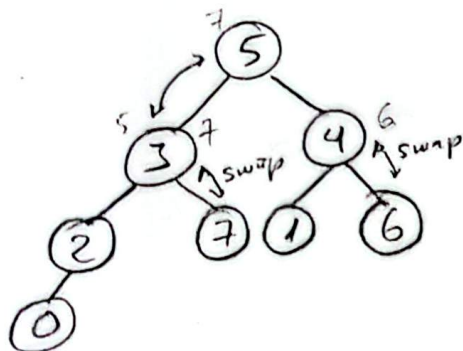
$$H_1 = \{5, 3, 4, 2\}$$

$$H_2 = \{7, 1, 6, 0\}$$

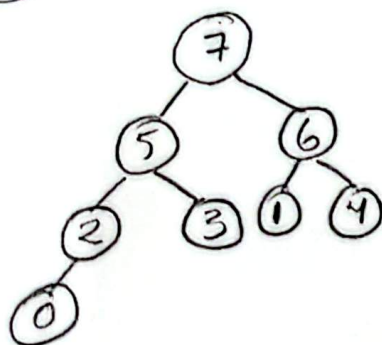
concatenate:

$$A = \{5, 3, 4, 2, 7, 1, 6, 0\}$$

buildMaxHeap using bottom-up



0	1	2	3	4	5	6	7	8
5	3	4	2	7	1	6	0	



0	1	2	3	4	5	6	7	8
3	2	4	7	1	6	0		

a) what is the best case?
 $O(n+n)$ due to concatenation and the loop over all parents.

b) What is the worst case?
 Still $O(n+m)$

c) Worst case time complexity?
 $O(n+m)$

In this case both heaps have same height
 $\Rightarrow n+m = n+n = 2n$

$$\underline{O(2n) = O(n)}$$